INCH-POUND

MIL-PRF-29597B (NAVY)
03 January 2000
SUPERSEDING
MIL-PRF-29597A (NAVY)
28 September 1998

PERFORMANCE SPECIFICATION

BAG, ODOR BARRIER, FLEXIBLE
(FOR FOOD CONTAMINATED PLASTIC WASTE)

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of odor barrier bags used to contain food-contaminated plastic waste (see 6.1).

1.2 Classification. Bags are furnished in the following types:

Type I – Pressure Sensitive Adhesive Closure
Type II – Heat Seal Closure

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in Sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in Sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Code 414100B120-3, Highway 547, Lakehurst, NJ 08733-5100, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

P-D-680	-	Dry Cleaning and Degreasing Solvent.

DEPARTMENT OF DEFENSE

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4, JP-5, and JP-5/JP-8 ST.
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR QUALITY CONTROL (ASQC)

ASQC-Z1.4	-	Procedures, Sampling and Tables for Inspection by Attributes. (DoD adopted)

(Applications for copies should be addressed to the American Society for Quality Control, P.O. Box 3005, 611 East Wisconsin Avenue, Milwaukee, WI 53201-4606.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D882	-	Plastic Sheeting, Tensile Properties of Thin. (DoD adopted)
ASTM-D898	-	Adhesive Solids, Dried, Applied Weight per Unit, Area of. (DoD adopted)
ASTM-D1922	-	Propagation Tear Resistance for Plastic Film and Thin Sheeting by Pendulum Method. (DoD adopted)

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ASTM-D3330	-	Tape, Pressure Sensitive, at 180 Deg. Angle, Peel Adhesion of. (DoD adopted)
ASTM-D3985	-	Standard Test Method for Oxygen Gas Transmission Rate Through Plastic Film Sheeting Using a Kilometric Sensor.
ASTM-F88	-	Seal Strength of Flexible Barrier Materials, Test Method for.
ASTM-F1249	-	Standard Test Method for Water-Vapor Transmission Rate Through Plastic Film and Sheeting Using a Modulated Infrared Sensor.

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), samples shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. The barrier material shall be made from such raw materials and by such processes as to ensure compliance with the requirements of this specification.

3.3 Form. The barrier material shall be converted into bags and furnished as bags.

3.3.1 Type I. Type I bag sizes (see 6.2) and inside dimensions when closed shall be as specified in table I with a tolerance of $\pm$ 0.5 inch (1.3 centimeters) for each dimension. The outside width of the bag after being closed shall be not greater than the dimensions specified in table I with a tolerance of + 1.5 inches (3.8 centimeters) for each. The bag shall be heat-sealed (see 3.9) on both sides with the bottom folded and creased. One side of the bag opening shall extend 2.25 to 2.75 inches (5.7 to 7.0 centimeters) higher than the other side of the bag opening creating a flap. A 4-inch-wide Pressure Sensitive Adhesive (PSA) strip shall be centered lengthwise over the top edge of the bag opening. As a result, both the inside of the flap and the outside of the lower bag opening shall have a 2 $\pm$ 0.125 inches (5 $\pm$ 0.32 centimeters) strip of PSA across the entire width of the bag. The PSA shall be covered with a release liner that extends past the edge (width) of the bag and shall be removable. The PSA strip and release liner shall not interfere with the opening of the bag prior to use.

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TABLE I. Type I and Type II bag sizes.

Size	Length (inches)	Width (inches)
A	50	36
B	27	24

3.3.2 Type II. Bags shall be created on the roll of lay flat tubing (see 3.3.2.1), by creating a heat-seal (see 3.9) every 50 + 0.25 inches (127 + 0.64 centimeters) for the size A bag, or 27 ± 0.25 inches (68 ± 0.64 centimeters) for the size B bag. Bag sizes (see 6.2) shall be as specified in table I. Perforations shall be created to enable bags to be removed from the roll by hand. The perforations shall occur below the heat-seal, and shall be less than 0.25 inch (0.64 centimeter) below the effective heat-seal. Perforations shall not interfere with the integrity of the seal (see 4.3.2).

3.3.2.1 Rolls. The average roll length shall be 75 ± 0.5 yards (69 ± 0.5 meters). The length of any individual roll shall be not less than 67 yards (61 meters). The width of roll material shall be 36 ± 0.25 inches (91 ± 0.64 centimeters) for size A bags or 24 ± 0.25 inches (61 ± 0.64 centimeters) for size B bags. The length of the core shall be equal to the width of roll material, with a tolerance of + 0.25 inch (0.64 centimeter). The core shall be rigid to prevent distortion of the roll during use and shipment. Each roll shall be restrained to prevent unwinding (see 4.3.2).

3.4 Thickness. Unless otherwise specified (see 6.2), the barrier material thickness shall be 0.005 inch (0.013 centimeter). When other thicknesses are required, they shall be specified in increments of 0.001 inch (0.003 centimeter). The tolerance shall be ± 5 percent of the specified thickness when measuring the average gage (see 4.3.2).

3.5 Material compatibility. The barrier material shall meet at least 75 percent of each mechanical property value specified in 3.7 and shall reveal no visual defects after being treated as specified in 4.4.1 with any of the following material:

a. Type II cleaning solvent in accordance with P-D-680, type II.

b. Hydraulic fluid in accordance with MIL-PRF-83282.

c. JP-5 grade kerosene in accordance with MIL-DTL-5624.

3.6 Permeability. The water vapor permeability shall be not greater than 5.97 grams per square meter per day for 0.005 inch (0.013 centimeter) thick barrier material when tested in accordance with 4.4.2. The oxygen permeability shall be not greater than 1.16 cubic centimeters (standard temperature and pressure) per square meter per day per atmosphere for 0.005 inch (0.013 centimeter) thick barrier material, when tested in accordance with 4.4.2.

3.7 Tensile strength, tear strength, and elongation. The barrier material tensile strength shall be not less than 4,910 pounds per square inch (33.8 mega pascals) in the machine direction and

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4,700 pounds per square inch (32.4 mega pascals) in the transverse direction, when tested in accordance with 4.4.3. The barrier material elongation shall be not less than 280 percent in the machine direction and 294 percent in the transverse direction, when tested in accordance with 4.4.3. The barrier material tear strength shall be not less then 96 grams-force in the machine direction and 124 grams-force in the transverse direction, when tested in accordance with 4.4.3.

3.8 Identification sheet. Each roll or package of bags shall contain an identification sheet with the specification number, manufacturer's name, manufacturer's designation, lot number, date of manufacture (month and year), national stock number, bag type, and size. Type II bags shall also contain sealing information; type of equipment, temperature, pressure, and dwell time (see 4.3.2).

3.9 Heat-seals. Heat-seals shall be 0.25 ± 0.05 inch (0.64 ± 0.13 centimeter). Seals shall be the flat impulse type whereby the barrier material is heated to its welding or melting temperature under pressure and allowed to cool while still under pressure. Heat-seals shall have a breaking strength that shall be not less than 15 pounds per inch (263 newtons per meter), when tested in accordance with 4.4.4.

3.10 Pressure Sensitive Adhesive Closure (type I only).

3.10.1 Peel adhesion. The laminating adhesive shall have a back side (liner side) adhesion to polypropylene that shall be not less than 48 ounces/inch of width when tested in accordance with 4.4.5.

3.10.2 Relaxation modulus ratio. The relaxation modulus ratio of the laminating adhesive shall be not less than 0.066 or greater than 0.160 when tested as specified in 4.4.6.

3.10.3 Laminating adhesive weight. The weight of the laminating adhesive shall be not less than 29 and not greater than 35 grains/24 square inches when tested in accordance with 4.4.7.

3.10.4 Material. The laminating adhesive shall be acrylic with no fibers or carrier within the adhesive (see 4.3.2).

3.11 Workmanship. The barrier material shall be free from any foreign matter. The barrier material edges shall be cut and trimmed of any selvage. The barrier material shall be free from holes, tears, cuts, wrinkles, or other imperfections (see 4.3.2).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

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b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall consist of all the tests and examinations specified in this specification.

4.3 Conformance inspection. Conformance inspections shall consist of the tests listed in table II and the examinations listed in 4.3.2.

TABLE II. Conformance tests.

Characteristics	Requirement paragraph	Test paragraph
Permeability	3.6	4.4.2
Tensile strength (only)	3.7	4.4.3
Heat-seal strength	3.9	4.4.4
Peel adhesion	3.10.1	4.4.5

4.3.1 Sampling for conformance inspection. For the purpose of determining the sample size in accordance with ASQC-Z1.4, the lot size shall be expressed in number of bags produced in one production run and shall use an inspection level of S-1 for testing and S-3 for visual examination.

4.3.2 Visual examination of end item for defects in form, thickness, identification, PSA material, and workmanship. The sample unit for the end item inspection shall be one bag. The sample unit shall be visually inspected and measured to ensure it meets the requirements specified in 3.3.2, 3.3.2.1, 3.4, 3.8, 3.10.4, and 3.11. A dial micrometer shall be used to measure the material thickness specified in 3.4.

4.4 Test methods. Unless otherwise specified all tests shall be conducted as specified herein. Barrier material and laminating adhesive samples shall be conditioned for 24 hours (minimum) in a circulating air environment having a relative humidity of 50 ± 5 percent and a temperature of 70 to 76 ºF (21 to 25 ºC).

4.4.1 Material compatibility. Barrier material samples shall be immersed in the chemicals (a separate specimen for each chemical) specified in 3.5 for 7 days at 68 ± 5 ºF (20 ± 3 ºC), and a visual examination after treatment shall be performed. The barrier material shall then be tested in accordance with 4.4.3.

4.4.2 Permeability. The water-vapor permeability shall be measured in accordance with ASTM-F1249. The oxygen permeability shall be tested in accordance with ASTM-D3985. Oxygen permeability shall be tested at less than 1 percent relative humidity.

4.4.3 Tensile strength, tear strength, and elongation. The barrier material tensile strength and elongation shall be tested in accordance with ASTM-D882. The barrier material tear strength shall be tested in accordance with ASTM-D1922.

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4.4.4 Heat-seal strength. The heat-seal strength shall be tested in accordance with ASTM-F88.

4.4.5 Peel adhesion test. The peel adhesion shall be tested in accordance with ASTM-D3330, Method B, using a 15 minute dwell on panel before peeling the masking tape backing.

4.4.6 Relaxation modulus ratio. The relaxation modulus G(t) ratio curve shall be obtained at 70 ºC and with a strain of 25 percent using a constant strain rheometer with parallel plate in shear mode. The relaxation modulus ratio shall be calculated by dividing relaxation modulus at 300 seconds by the relaxation modulus at 0.1 second.

4.4.7 Laminating adhesive weight. The laminating adhesive weight shall be tested in accordance with ASTM-D898.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points packaging activity within the Military Department or Defense Agency, or within The Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The bags provide an odor barrier for the containment of food-contaminated plastic waste in shipboard applications where at-sea disposal of plastics is prohibited. There are no commercial equivalents that meet all of the physical and mechanical requirements necessary for this application. In use, filled bags are intended to remain aboard ship for prolonged periods of time in consonance with naval deployments.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of the DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

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c. Bag type (see 1.2).

d. Bag size (see table I).

e. Thickness (if other than 0.005 inch) (see 3.4).

f. Details for first article inspections/testing (see 4.2 and 6.3).

g. Packaging requirements (see 5.1).

6.3 First article. The contracting officer should provide specific guidance to offerors whether the first article samples should be a preproduction item, a first production item, a sample selected from the first production items or a standard production item from the contractor's current inventory (see 3.1 and 4.2). The contracting officer should also include specific instructions in acquisition documents regarding arrangements for submission of Material Safety Data Sheets (MSDS) prior to contract award, testing, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive requirements for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government. Bidders offering such products, that wish to rely on such production or test, must furnish evidence with the bid that the prior Government approval is acceptable for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.3.1 Prior approval. If a contractor has previously delivered an acceptable product meeting the requirements of this specification, waiver of the first article inspection requirement (see 4.2) is at the discretion of the procuring activity for a period of time not to exceed 2 years (see 6.2).

6.3.2 First article sample. The first article sample should be produced using the same plant facilities intended for the production run. Failure of the first article sample to meet all the requirements of this specification is cause for rejection.

6.4 Subject term (keyword) listing.

Barrier material
Heat-seal
Permeability
Plastic film
Oxygen gas transmission rate
Waste bag
Watervapor transmission rate

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodian:
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Review activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-29597B(NAVY)	**2. DOCUMENT DATE** (*YYYYMMDD*) 20000103

3. DOCUMENT TITLE
BAG, ODOR BARRIER, FLEXIBLE (FOR FOOD CONTAMINATED PLASTIC WASTE)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include ZIP Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) DSN (*If applicable*)	**7. DATE SUBMITTED** (*YYYYMMDD*)

8. PREPARING ACTIVITY

a. NAME COMMANDER NAVAL AIR WARFARE CENTER AIRCRAFT DIVISION	b. TELEPHONE (*Include Area Code*) (1) Commercial (732) 323-2628 (2) DSN 624-2628
c. ADDRESS (*Include ZIP Code*) CODE 414100B120-3 HIGHWAY 547 LAKEHURST, NJ 08733-5100	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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